Assignment 3: Mathematical Formulae in LaTeX

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Some Important Mathematical Formulae

Geometry

Circle : Area = πr^2; Circumference = $2\pi r$.
Square : Area = x^2; Perimeter = $4x$.
Rectangle : Area = xy; Perimeter = 2(x+y).
Triangle : Area = $\frac{1}{2} \times$ base $\times$ height; Perimeter = $a + b + c$.
Area of equilateral triangle = $\frac{\sqrt{3}}{4}a^2$.

Sphere : Surface Area = $4\pi r^2$; Volume = $\frac{4}{3}\pi r^3$.
Cube : Surface Area = $6a^2$; Volume = a^3.
Cone : Curved Surface Area = πrl; Volume = $\frac{1}{3}\pi r^2 h$
Total surface area = $\pi rl + \pi r^2$.
Cuboid : Total surface area = 2(ab+bh+lh); Volume = lbh.
Cylinder : Curved surface area = $2\pi rh$; Volume = $\pi r^2 h$
Total surface area (open) = $2\pi rh$;
Total surface area (closed) = $2\pi rh + 2\pi r^2$.

Algebraic Formulae

1. $(a + b)^2 = a^2 + 2ab + b^2$.
2. $(a - b)^2 = a^2 - 2ab + b^2$.
3. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$.
4. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$.
5. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.
6. $(a+b+c)^3 = a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3b^2c + 3b^2a + 3c^2a + 3c^2a + 6abc$.
7. $a^2 - b^2 = (a + b)(a - b)$.
8. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

9. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$.
10. $(a + b)^2 + (a - b)^2 = 4ab$.
11. $(a + b)^2 - (a - b)^2 = 2(a^2 + b^2)$.
12. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.

Indices and Surds

1. $a^m a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $(ab)^m = a^m b^m$
5. $(\frac{a}{b})^m = \frac{a^m}{b^m}$
6. $a^0 = 1, a \neq 0$.
7. $a^{-m} = \frac{1}{a^m}$
8. $a^x = a^y \Rightarrow x = y$
9. $a^x = b^x \Rightarrow a = b$
10. $\sqrt{a \pm 2\sqrt{b}} = \sqrt{x} \pm \sqrt{y}$, where $x + y = a$ and $xy = b$.

Logarithms

$a^x = m \Rightarrow \log_a m = x (a > 0 \; and \; a \neq 1)$

1. $\log_a mn = \log m + \log n$.
2. $\log(\frac{m}{n}) = \log m - \log n$
3. $\log_a m^n = n \;\; \log m$.
4. $\log_b a = \frac{\log a}{\log b}$
5. $\log_a a = 1$
6. $\log_a 1 = 0$
7. $\log_b a = \frac{1}{\log_a b}$
8. $\log_a 1 = 0$.
9. $\log(m + n) \neq \log m + \log n$
10. $e^{\log x} = x$.
11. $\log_a a^x = x$.

Progressions

Arithmetic Progression

$a, a+d, a+2d, \ldots$ are in A.P n^{th} term, $T_n = a + (n-1)d$
Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$ If a, b, c are in A.P, then $2b = a + c$.

Geometric Progression

$a, ar, ar^2, \ldots$ are in G.P Sum to n terms, $S_n = \frac{a(1-r^n)}{1-r}$ if $r < 1$ $S_n = \frac{a(r^n-1)}{r-1}$ if $r > 1$. Sum to infinite terms of G.P, $S_\infty = \frac{a}{1-r}$ If a, b, c are in G.P, then $b^2 = ac$.

Harmonic Progression

Reciprocals of the terms of A.P are in H.P $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \ldots$ are in H.P If a, b, c are in H.P, then $b = \frac{2ac}{a+c}$

Mathematical Induction

$$1 + 2 + 3 + \cdots + n = \sum n = \frac{n(n+1)}{2}$$
$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$
$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

Permutations and Combination

$nP_r = \frac{n!}{(n-r)!}$
$nC_r = \frac{n!}{r!(n-r)!}$
$n! = 1 \cdot 2 \cdot 3 \ldots n.$
$nC_r = nC_{n-r}.$
$nC_r + nC_{r-1} = (n+1)C_r$
$(m+n)C_r = \frac{(m+n)!}{m!n!}$

Binomial Theorem

$(x+a)^n = x^n + nC_1 x^{n-1}a + nC_2 x^{n-2}a^2 + nC_3 x^{n-3}a^3 + \cdots + nC_n a^n$ n^{th} term, $T_{r+1} = nC_r x^{n-r}a^r$.

Partial Fractions

$\frac{f(x)}{g(x)}$ is a proper fraction if the $\deg(g(x)) > \deg(f(x))$.
$\frac{f(x)}{g(x)}$ is an improper fraction if the $\deg(g(x)) \leq \deg(f(x))$.

1. Linear non-repeated factors: $\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{(cx+d)}$
2. Linear repeated factors: $\frac{f(x)}{(ax+b)(cx+d)^2} = \frac{A}{ax+b} + \frac{B}{(cx+d)} + \frac{C}{(cx+d)^2}$
3. Non-linear (quadratic which can not be factorized): $\frac{f(x)}{(ax^2+b)(cx^2+d)} = \frac{Ax+B}{ax^2+b} + \frac{Cx+D}{(cx^2+d)}$

Analytical Geometry

1. Distance between the two points (x_1, y_1) and (x_2, y_2) in the plane is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ OR $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$.
2. Section formula:
   $(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n})$ (for internal division)
   $(\frac{mx_2-nx_1}{m-n}, \frac{my_2-ny_1}{m-n})$ (for external division).
3. Mid point formula: $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.
4. Centroid formula: $(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3})$.
5. Area of triangle when their vertices are given: $\frac{1}{2}\sum x_1(y_2-y_3) = \frac{1}{2}[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)]$

Straight Line

Slope (or Gradient) of a line = tangent of an inclination = $\tan\theta$.
Slope of a X- axis = 0
Slope of a line parallel to X-axis = 0
Slope of a Y- axis = ∞
Slope of a line parallel to Y-axis = ∞
Slope of a line joining (x_1, x_2) and (y_1, y_2) = $\frac{y_2-y_1}{x_2-x_1}$
If two lines are parallel, then their slopes are equal ($m_1 = m_2$)
If two lines are perpendicular, then their product of slopes is -1 ($m_1 m_2 = -1$).

Equations of Straight Line

1. $y = mx + c$ (slope-intercept form)
   $y - y_1 = m(x - x_1)$ (point-slope form)
   $y - y_1 = \frac{y_2-y_1}{x_2-x_1}(x - x_1)$ (two point form)
   $\frac{x}{a} + \frac{y}{b} = 1$ (intercept form)
   $x\cos\alpha + y\sin\alpha = P$ (normal form)
   Equation of a straight line in the general form is $ax^2 + bx + c = 0$.
   Slope of $ax^2 + bx + c = 0$ is $-(\frac{a}{b})$.
2. Angle between two straight lines is given by, $\tan\theta = |\frac{m_1-m_2}{1+m_1m_2}|$
   Length of the perpendicular from a point (x_1, y_1) to the straight line $ax +$

$by + c = 0$ is

$$|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}|$$

Equation of a straight line passing through intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is $a_1x + b_1y + c_1 + K(a_2x + b_2y + c_2) = 0$ where K is any constant.
Two lines meeting a point are called intersecting lines.
More than two lines meeting a point are called concurrent lines.
Equation of bisector of angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is

$$\frac{a_1x + b_1y + c_1}{\sqrt{{a_1}^2 + {b_1}^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{{a_2}^2 + {b_2}^2}}$$

Pair of Straight Lines

1. An equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin. The angle between these is given by

   $$\tan\theta = |\frac{2\sqrt{h^2 - ab}}{a + b}|$$

   $ax^2 + 2hxy + by^2 = 0$ represents a pair of coincident lines, if $h^2 = ab$ and the same represents a pair of perpendicular lines, if $a + b = 0$. If m_1 and m_2 are the slopes of the lines $ax^2 + 2hxy + by^2 = 0$ then $m_1 + m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$

2. An equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is called the general second order equation. It represents a pair of lines if it satisfies the condition

   $$abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

   The angle between the lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is given by

   $$\tan\theta = |\frac{2\sqrt{h^2 - ab}}{a + b}|$$

   $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines, if $h^2 = ab$ and $af^2 = bg^2$; and the distance between the parallel lines is

   $$\frac{2\sqrt{g^2 - ac}}{a(a + b)}$$

   $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of perpendicular lines, if $a + b = 0$.

Trignometry

Area of a sector of a circle $= \frac{1}{2}r^2\theta$. Arc length, $S = r\theta$ $sin\theta = \frac{opp}{hyp}$, $\cos\theta = \frac{adj}{hyp}$ $\tan\theta = \frac{opp}{adj}$ $\cot\theta = \frac{adj}{opp}$ $\sec\theta = \frac{hyp}{adj}$ $\csc\theta = \frac{hyp}{opp}$. $\sin\theta = \frac{1}{\csc\theta}$ or $\csc\theta = \frac{1}{\sin\theta}$ $\cos\theta = \frac{1}{\sec\theta}$ or $\sec\theta = \frac{1}{\cos\theta}$, $\tan\theta = \frac{1}{\cot\theta}$ or $\cot\theta = \frac{1}{\tan\theta}$, $\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\cot\theta = \frac{\cos\theta}{\sin\theta}$. $\sin^2\theta + \cos^2\theta = 1$;$\Rightarrow \sin^2\theta = 1 - \cos^2\theta$; $\cos^2\theta = 1 - \sin^2\theta$ $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec^2\theta = 1 + \tan^2\theta$; $\tan^2\theta = \sec^2\theta - 1$; $\csc^2\theta - \cot^2\theta = 1$;$\Rightarrow \csc^2\theta = 1 + \cot^2\theta$ $\cot^2\theta = \csc^2\theta - 1$

Allied Angles

If the transformation begins at 90° or 270^0, the trigonometric functions change as
$\sin \leftrightarrow \cos$
$\tan \leftrightarrow \cot$
$\sec \leftrightarrow \csc$.
Whereas the transformation begins at 180° or 360°, the same trigonometric functions will be retained.

Compound Angles

$\sin(A + B) = \sin A\cos B + \cos A\sin B$.
$\sin(A - B) = \sin A\cos B - \cos A\sin B$.
$\cos(A + B) = \cos A\cos B - \sin A\sin B$
$\cos(A - B) = \cos A\cos B + \sin A\sin B$.
$\tan(A + B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$
$\tan(A - B) = \frac{\tan A-\tan B}{1+\tan A\tan B}$
$\tan(\frac{\pi}{4} + A) = \frac{1+\tan A}{1-\tan A}$
$\tan(\frac{\pi}{4} - A) = \frac{1-\tan A}{1+\tan A}$
$\tan(A + B + C) = \frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-(\tan A\tan B+\tan B\tan C+\tan C\tan A)}$
$\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
$\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$

Multiple Angles

1. $\sin 2A = 2\sin A\cos A$.
2. $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$
3. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1 = \frac{1-\tan^2 A}{1+\tan^2 A}$
4. $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$
5. $1 + \cos 2A = 2\cos^2 A$,
6. $\cos^2 A = \frac{1}{2}(1 + \cos 2A)$.

7. $1 - \cos 2A = 2\sin^2 A$
8. $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$.
9. $1 + \sin 2A = (\sin A + \cos A)^2$,
10. $1 - \sin 2A = (\cos A - \sin A)^2 = (\sin A - \cos A)^2$,
11. $\cos 3A = 4\cos^3 A - 3\cos A$
12. $\sin 3A = 3\sin A - 4\sin^3 A$,
13. $\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Half Angle Formulae

1. $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$.
2. $\sin\theta = \frac{2\tan(\frac{\theta}{2})}{1+\tan^2(\frac{\theta}{2})}$.
3. $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$
4. $\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$
5. $\cos\theta = 2\cos^2\frac{\theta}{2} - 1$
6. $\cos\theta = \frac{1-\tan^2(\frac{\theta}{2})}{1+\tan^2(\frac{\theta}{2})}$.
7. $\tan\theta = \frac{2\tan(\frac{\theta}{2})}{1-\tan^2(\frac{\theta}{2})}$.
8. $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$
9. $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$

Product to Sum

$2\sin A\cos B = \sin(A+B) + \sin(A-B)$.
$2\cos A\sin B = \sin(A+B) - \sin(A-B)$.
$2\cos A\cos B = \cos(A+B) + \cos(A-B)$.
$2\sin A\sin B = \cos(A+B) - \cos(A-B)$.

Sum to Product

$\sin C + \sin D = 2\sin(\frac{C+D}{2})\cos(\frac{C-D}{2})$.
$\sin C - \sin D = 2\cos(\frac{C+D}{2})\sin(\frac{C-D}{2})$.
$\cos C + \cos D = 2\cos(\frac{C+D}{2})\cos(\frac{C-D}{2})$.
$\cos C - \cos D = -2\sin(\frac{C+D}{2})\sin(\frac{C-D}{2})$
OR
$\cos C - \cos D = 2\sin(\frac{D+C}{2})\sin(\frac{D-C}{2})$.

Properties and Solutions of Triangle

Sine Rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$ where R is the circum radius of the triangle.

Cosine Rule: $a^2 = b^2 + c^2 - 2bc\cos A$ or $\cos A = \frac{b^2+c^2-a^2}{2bc}$.

$b^2 = a^2 + c^2 - 2ac\cos B$ or $\cos B = \frac{a^2+c^2-b^2}{2ac}$.

$c^2 = a^2 + b^2 - 2ab\cos C$ or $\cos C = \frac{a^2+b^2-c^2}{2ab}$.

Projection Rule: $a = b\cos C + c\cos B$

$b = c\cos A + a\cos C$

$c = a\cos B + b\cos A$

Tangents Rule: $\tan(\frac{B-C}{2}) = \frac{b-c}{b+c}\cot(\frac{A}{2})$,

$\tan(\frac{C-A}{2}) = \frac{c-a}{c+a}\cot(\frac{B}{2})$,

$\tan(\frac{A-B}{2}) = \frac{a-b}{a+b}\cot(\frac{C}{2})$.

Half angle formula: $\sin(\frac{A}{2}) = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\cos(\frac{A}{2}) = \sqrt{\frac{s(s-a)}{bc}}$, $\tan(\frac{A}{2}) = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$.

$\sin(\frac{B}{2}) = \sqrt{\frac{(s-a)(s-c)}{ac}}$, $\cos(\frac{B}{2}) = \sqrt{\frac{s(s-b)}{ac}}$, $\tan(\frac{B}{2}) = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$.

$\sin(\frac{C}{2}) = \sqrt{\frac{(s-a)(s-b)}{ab}}$, $\cos(\frac{C}{2}) = \sqrt{\frac{s(s-c)}{ab}}$, $\tan(\frac{C}{2}) = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$.

Area of triangle $ABC = \sqrt{s(s-a)(s-b)(s-c)}$

Area of triangle $ABC = \frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B = \frac{1}{2}ab\sin C$

Limits

1. If $f(-x) = f(x)$, then $f(x)$ is called Even Function.
2. If $f(-x) = -f(x)$, then $f(x)$ is called Odd Function.
3. If P is the smallest +ve real number such that if $f(x+P) = f(x)$, then $f(x)$ is called a periodic function with period P.
4. Right Hand Limit (RHL) $= \lim_{x\to a^+}(f(x)) = \lim_{h\to 0}(f(a+h))$
   Left Hand Limit (LHL) $= \lim_{x\to a^-}(f(x)) = \lim_{h\to 0}(f(a-h))$
   If $RHL = LHL$ then $\lim_{x\to a}(f(x))$ exists and $\lim_{x\to a}(f(x)) = RHL = LHL$
5. $Lt_{n\to\infty}\frac{1}{n^p} = 0$ if $p > 0$ and $Lt_{n\to\infty}n^p = \infty$ if $p > 0$
6. $Lt_{x\to 0}\frac{\sin x}{x} = Lt_{x\to 0}\frac{\tan x}{x}$ (x in radians) $= Lt_{x\to 0}\frac{x}{\sin x} = Lt_{x\to 0}\frac{x}{\tan x} = 1$
7. $Lt_{x\to 0}\frac{\sin x^0}{x} = Lt_{x\to 0}\frac{\tan x^0}{x} = \frac{\pi}{180}$
8. $Lt_{x\to\frac{\pi}{2}}\frac{\sin x}{x} = \frac{2}{\pi}$
9. $\lim_{x\to 0}\frac{\sin^{-1}x}{x} = 1 = \lim_{x\to 0}\frac{\tan^{-1}x}{x}$
10. $\lim_{x\to a}\frac{x^n-a^n}{x-a} = na^{n-1}$, where n is an integer or a fraction.

11. $\lim_{x\to 0} \frac{a^x-1}{x} = \log a$, $\lim_{x\to 0} \frac{e^x-1}{x} = \log e = 1$

12. $\lim_{n\to\infty}(1+\frac{1}{n})^n = e$, $\lim_{x\to 0}(1+n)^{\frac{1}{n}} = e$

13. $\lim_{x\to a}[kf(x)] = k\lim_{x\to a} f(x)$

14. $\lim_{x\to a}[f(x) \pm g(x)] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$

15. $\lim_{x\to a} f(x).g(x) = \lim_{x\to a} f(x).\lim_{x\to a} g(x)$
$\lim_{x\to a}(\frac{f(x)}{g(x)}) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ provided $\lim_{x\to a} g(x) \neq 0$

16. A function $f(x)$ is said to be continuous at the point $x = a$ if
    (a) $\lim_{x\to a} f(x)$ exists
    (b) $f(a)$ is defined
    (c) $\lim_{x\to a} f(x) = f(a)$

17. A function $f(x)$ is said to be discontinuous or not continuous at $x = a$ if
    (a) $f(x)$ is not defined at $x = a$
    (b) $\lim_{x\to a} f(x)$ does not exist at $x = a$
    (c) $\lim_{x\to a+0} f(x) \neq \lim_{x\to a-0} f(x) \neq f(a)$

18. If two functions $f(x)$ and $g(x)$ are continuous then $f(x) + g(x)$ is continuous.